

Deep Thought Culture Tech — Task 1 Submission

Role: Clinic Business Analyst — Daily Clinic Inventory Micro-Audit

Objective

Design a daily system that detects inventory and sales-entry errors, improves staff compliance, and prevents month-end shocks. The system should run in 20–30 minutes per day and involve the doctor only when necessary.

STEP 1 — Identify High-Risk Medicines (One-Time Setup)

Purpose: Focus daily checks on medicines most likely to have errors.

Criteria for selecting high-risk medicines:

1. High daily sales volume
2. Similar or confusing names
3. Frequently prescribed by doctor
4. High value per unit

High-Risk Medicines Table:

Medicine	Why High Risk
Dolo 650	High volume + name variations
Azithromycin 500	Similar abbreviations used
Pantoprazole 40	Frequently prescribed

Time: ~30 mins (one-time setup)

STEP 2 — Daily Name-Variation Check (10 mins/day)

Purpose: Detect sales-entry mistakes early.

Process:

1. Filter Sales Register for high-risk medicines only.
2. Group similar names manually or using text similarity.

3. Compare with Master Name.

Example Name Mapping Table:

Entered Name Variants	Mapped To
Dolo / Dolo 650 / Dolo kind	Dolo 650
Azithro 500 / Azithromycin	Azithromycin 500

Flag / Action:

- Flag new or unusual variants
- Add to mapping list
- Inform billing staff to use standard names

STEP 3 — Daily Usage Reasonableness Check

Purpose: Compare expected vs actual closing stock to detect quantity errors.

Formula:

Expected Closing Stock = Opening + Purchases – Total Sales

Sample Table:

Medicine	Expected	Actual	Difference	Action
Dolo 650	1460	1440	-20	Review sales entries
Azithromycin 500	500	480	-20	Review sales entries

Tolerance: $\pm 2\%$

Flag action: Investigate differences beyond tolerance. Escalate to doctor only if material.

STEP 4 — Random Bill Spot Check (5–10 mins/day)

Purpose: Ensure staff accuracy with manual sales entries.

Process:

1. Randomly pick 3–5 bills involving high-risk medicines.
2. Verify:
 - Medicine name matches Master Name
 - Quantity matches prescription / sales record

Rules:

- No punishment, only correction
- Random checks improve staff attention naturally

STEP 5 — Pattern Tracking (Weekly, 15 mins)

Purpose: Detect recurring errors and improve SOPs

Error Log Table:

Date	Medicine	Error Type	Repeated?
Aug 1	Dolo 650	Name variant	Yes
Aug 3	Dolo 650	Wrong quantity	No

Decision Rules:

- Same error ≥ 3 times → retrain staff
- Multiple medicines affected → update SOP

STEP 6 — Doctor Escalation Rules**Escalate only if:**

- Financial impact is material
- Pattern persists despite correction
- Inventory mismatch affects treatment availability

Everything else is handled by **analyst/staff**, not the doctor.

Daily Checklist (20–30 mins/day)

1. Name-Variation Check for High-Risk Medicines (10 mins)
2. Usage Reasonableness Check (10 mins)
3. Random Bill Spot Check (5–10 mins)

Weekly Checklist (15 mins/week)

1. Review Error Log
2. Identify recurring issues
3. Retrain staff or update SOPs as needed

Summary:

This system is designed to catch errors early, change staff behaviour, and protect doctor time, while being practical enough to run every day in a small clinic.